ANIKET RAI

Senior Software Engineer

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SUMMARY

Senior Software Engineer with 5+ years of experience building scalable backend systems and data pipelines for a B2B SaaS cloud-backup platform. Deep expertise in Node.js, TypeScript, and PostgreSQL (advanced query optimization, CTEs, materialized views). Experienced in owning large-scale framework migrations end-to-end, designing config-driven architectures, and shipping high-impact features across the full stack. Recently built AI agents for automated database monitoring and operational workflows, and an AI-powered natural language interface that lets customers query and act on their data without manual navigation.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript (Node.js), SQL, Python
Backend & APIs:	Node.js, Express.js, GraphQL (PostGraphile), REST APIs, Microservices, Batch Processing, Config-Driven Architecture
Databases:	PostgreSQL (CTEs, Recursive CTEs, Materialized Views, Functions, Views, Query Optimization), AWS Athena, MongoDB
Cloud & Infrastructure:	AWS (S3, Lambda, Fargate, Batch, SQS, CodeCommit, CodeBuild, CodePipeline, S3 Event Notifications, EventBridge), Docker, Serverless
Monitoring & Observability:	Grafana, ELK Stack (Elasticsearch, Logstash, Kibana)
AI & Automation:	AI Agent Development, LLM Integration, Claude, MCP, Anomaly Detection
Tools & Practices:	Git, CI/CD (AWS CodePipeline), Unit Testing, Integration Testing, SonarQube, Azure DevOps, System Design, Code Reviews, Agile

WORK EXPERIENCE

SysCloud Technologies Pvt Ltd

Senior Software Engineer — Hyderabad, Telangana | Jun 2023 – Present

- Revamped the **customer-facing Dashboard** to display aggregated backup data across 12+ clouds (License Management, Restore, Export, Add-on services) with zero code changes per new cloud, as all queries and aggregations are dynamically driven from a metadata table; also launched the **Partner Portal end-to-end** to onboard MSPs onto the platform.
- Delivered the **Restore & Export action layer for S3 Data Migration** where older metadata (2+ years) lives in S3; the action fetches it on-demand via AWS Athena and populates the DB, reducing Archives query time from 10s+ to under 2s for large datasets.
- Integrated **Slack eDiscovery Search** support, enabling customers to search and retrieve their Slack files, messages, and chats through the product's eDiscovery interface.
- Launched **MDLOPS, a new TypeScript microservice from scratch,** using cm-runner-plus + JSON-driven config to handle 5 add-on modules (eDiscovery, Hold, Data Change Insights, Archiver, BDI) with parallel execution managed by PM2; integrated anomaly detection (Changed, Added, Deleted file tracking) using the **Prophet algorithm** for trend-level analysis.
- Built **4 DAL AI-Agents** (Database Health, Restore/Export, StartBackup, PostGraphile Slow-Queries) to automate operational monitoring and detect and resolve issues automatically before they impact customers.
- Designed and owned **MDL 2.0,** a framework that replaced complex function/mutation logic with simple JSON-driven configuration, executing each action as a micro-task inside the PostGraphile server; migrated 12+ cloud integrations (Restore/Export, Sync, OnDemand Backup, Access Management, special apps) to MDL 2.0, so adding a new cloud now requires zero code changes, only config.

- Implemented **Sky 2.0 (SysCloud AI),** an AI-powered natural language interface where customers ask questions and the AI dynamically builds queries, redirects to the relevant page with pre-applied filters, and initiates Restore & Export actions, eliminating manual browsing through Archives.
- Documented all platform tables, views, and functions as structured AI-agent context for Sky 2.0, enabling accurate automated query generation across the entire product.
- Overhauled the **Batch Job & Materialized View refresh framework** by introducing parallel host-level execution, achieving 4x faster job execution; redesigned MVW refresh (login, on-demand, scheduled) to execute in transactional DB first then sync to cache DB via foreign keys, and added a centralized job table for all module-level refresh jobs.
- Modeled **parent/child schema and mutation layer** for Add-On Actions (eDiscovery, BDI, common workflows) including Dismiss, Mark True-Positive, Hold/Release, Archive On-Demand, Remove Sharing, Transfer Ownership, and Deletion.
- Shipped **Gen4 & Gen5 cloud pages** (Reports, Restore, Export, Dashboard, Unified Errors, Limitations) and Homepage widgets (Audit Log, ETC, Errors & Limitations, License Management, Usage Stats) with redirection to Search-Aggregated and leaf pages; introduced a **config-driven template** so any new cloud onboarded to the platform is automatically supported across all pages without additional code changes.
- Resolved **SonarQube code quality issues** across the codebase, eliminating all critical security violations (reduced to 0) and cutting total issues by 90%.

Software Engineer Chennai, Tamil Nadu | May 2021 – Jun 2023

- Architected a **unified config-driven Restore/Export function** (Gen3) that eliminated the need for code changes when onboarding new cloud providers, and introduced CM Download Status + ETA calculation.
- Developed a **Timeseries and Batch Framework** with multi-environment version support that schedules and executes UI Materialized View (MVW) refresh jobs on a schedule and on login, ensuring customers always see up-to-date data when they open the app; applied CPU-level optimizations and function-call tuning for improved throughput.
- Engineered a **weekly S3 data integrity pipeline** using Node.js, AWS Fargate, Lambda, and Athena to detect two classes of backup issues: (1) zip files entirely missing from S3, and (2) zip files with mismatched file counts vs. database records, automatically triggering re-backup for both, with Grafana monitoring for visibility.
- Integrated **ELK Stack with Grafana** for centralized backend error log collection and real-time CM error monitoring across the application framework.
- Introduced a **version-copy mechanism** to identify S3 objects with multiple versions, extract older ones, and create new objects, resolving restoration failures caused by outdated file associations in the DB.

Software Engineer Intern Chennai, Tamil Nadu | Nov 2020 – Apr 2021

- Authored GraphQL (GQL) queries for UI modules and implemented functions for mutation calls within the PostGraphile server.
- Explored AWS services and delivered multiple POCs that contributed to the company's architectural shift from a monolithic system (Gen1) to multiple microservices (Gen2).

EDUCATION

Bachelor of Technology – Computer Science & Engineering (B.Tech CSE)
Lovely Professional University | 2017 – 2021

CERTIFICATIONS

- Node.js Certificate Training – Simplilearn
- Full Stack Development – upGrad
- ChatGPT & AI Tools Workshop – Be10x